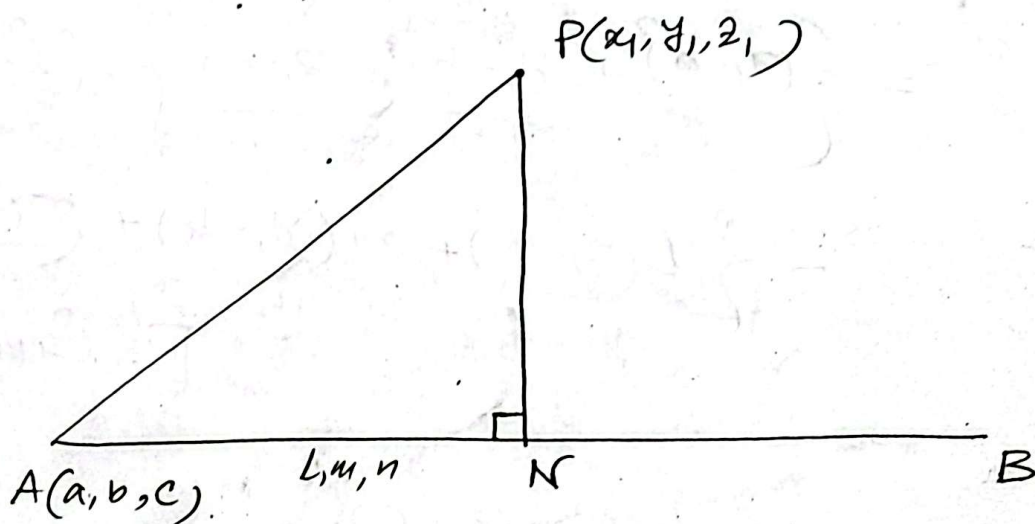


Th^mo

Find the perpendicular distance of a point $P(x_1, y_1, z_1)$ from a line through the point $A(a, b, c)$ and whose direction cosines are l, m, n .

Solⁿ:



Let AB be a line through $A(a, b, c)$ and $P(x_1, y_1, z_1)$ be a point. Draw $PN \perp AB$ from P ; join AP .

Now,

$AN = \text{Projection of } AP \text{ upon } AB.$

$$= l(x_1 - a) + m(y_1 - b) + n(z_1 - c)$$

$AP = \text{Distance bet}^n \text{ the two points } A \text{ and } P$

$$= \sqrt{(x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2}$$

$$\Rightarrow AP^2 = (x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2$$

From ΔAPN , using Pythagoras th^m we get

$$AP^2 = AN^2 + PN^2$$

$$\Rightarrow PN^2 = AP^2 - AN^2$$

$$\begin{aligned} \Rightarrow PN^2 &= \left\{ (x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2 \right\} - \left\{ l(x_1 - a) + m(y_1 - b) + n(z_1 - c) \right\}^2 \\ &= \left\{ \frac{(x_1 - a)^2}{a_1} + \frac{(y_1 - b)^2}{b_1} + \frac{(z_1 - c)^2}{c_1} \right\} \cdot \left(\frac{l^2}{a_2} + \frac{m^2}{b_2} + \frac{n^2}{c_2} \right) \\ &\quad - \left\{ \frac{l(x_1 - a)}{a_1 a_2} + \frac{m(y_1 - b)}{b_1 b_2} + \frac{n(z_1 - c)}{c_1 c_2} \right\}^2 \\ &\quad [\because l^2 + m^2 + n^2 = 1] \end{aligned}$$

$$\begin{aligned} &= \left\{ n(y_1 - b) - m(z_1 - c) \right\}^2 \\ &\quad + \left\{ l(z_1 - c) - n(x_1 - a) \right\}^2 \\ &\quad + \left\{ m(x_1 - a) - l(y_1 - b) \right\}^2 \end{aligned}$$

[using Lagrange's Identity]

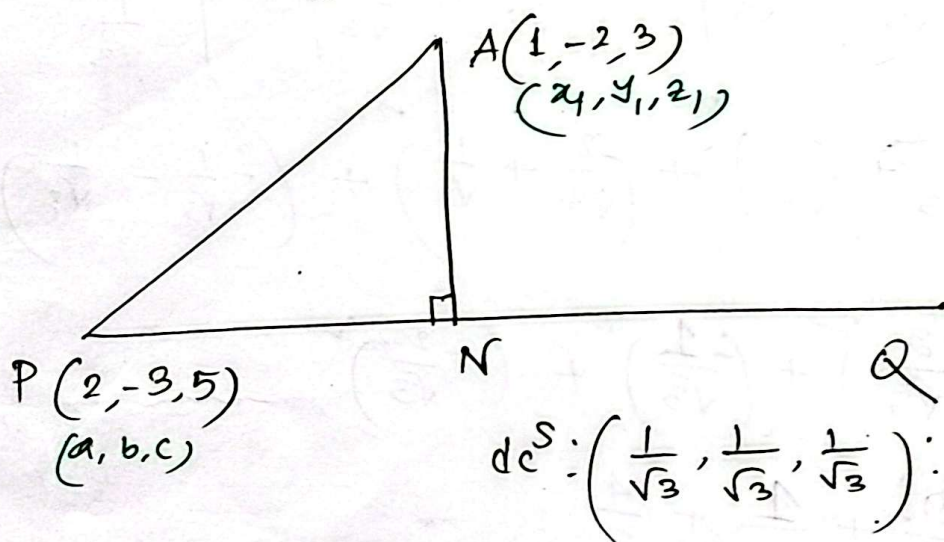
$$\therefore PN^2 = \begin{vmatrix} y_1 - b & z_1 - c \\ m & n \end{vmatrix}^2 + \begin{vmatrix} z_1 - c & x_1 - a \\ n & l \end{vmatrix}^2 + \begin{vmatrix} x_1 - a & y_1 - b \\ l & m \end{vmatrix}^2$$

Problem: (Incourse 2022)

Find the distance of $A(1, -2, 3)$ from the line PQ through $P(2, -3, 5)$ which makes equal angles with the axes.

Solⁿ:

Let l, m, n be the direction cosines of the line PQ . Since, PQ makes equal angle α with x -axis, y -axis and z -axis, therefore $l = m = n$ ($\cos \alpha = \cos \beta = \cos \gamma$)



But, $l^2 + m^2 + n^2 = 1$

$\Rightarrow 3l^2 = 1 \Rightarrow l = \frac{1}{\sqrt{3}}$; Similarly, $m = \frac{1}{\sqrt{3}}$
& $n = \frac{1}{\sqrt{3}}$

$\therefore l = m = n = \frac{1}{\sqrt{3}}$

If AN be the required perpendicular distance of A from the line PQ , then we have

$$AN^2 = \left| \begin{array}{cc} y_1 - b & x_1 - c \\ m & n \end{array} \right|^2 + \left| \begin{array}{cc} x_1 - c & x_1 - a \\ n & l \end{array} \right|^2 + \left| \begin{array}{cc} x_1 - a & y_1 - b \\ l & m \end{array} \right|^2$$

$$= \left| \begin{array}{cc} -2+3 & 3-5 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{array} \right|^2 + \left| \begin{array}{cc} 3-5 & 1-2 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{array} \right|^2 + \left| \begin{array}{cc} 1-2 & -2+3 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{array} \right|^2$$

$$= \left| \begin{array}{cc} 1 & -2 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{array} \right|^2 + \left| \begin{array}{cc} -2 & -1 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{array} \right|^2 + \left| \begin{array}{cc} -1 & 1 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{array} \right|^2$$

$$= \left(\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} \right)^2 + \left(\frac{-2}{\sqrt{3}} + \frac{1}{\sqrt{3}} \right)^2 + \left(-\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{3}} \right)^2$$

$$= \left(\frac{3}{\sqrt{3}} \right)^2 + \left(\frac{-1}{\sqrt{3}} \right)^2 + \left(-\frac{2}{\sqrt{3}} \right)^2$$

$$= \frac{9}{3} + \frac{1}{3} + \frac{4}{3}$$

$$= \frac{14}{3}$$

$$\Rightarrow AN = \sqrt{\frac{14}{3}}$$

$$= \frac{\sqrt{14}}{\sqrt{3}}$$

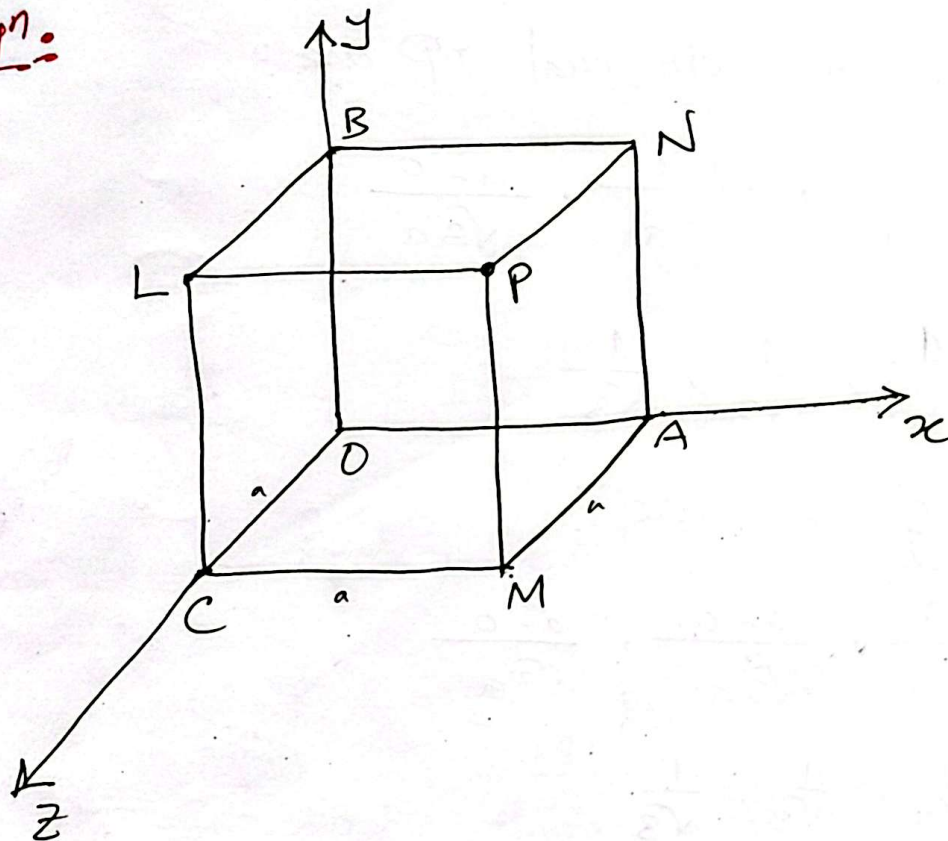
Ans)

Date: 9/9/22

Problem:

A line makes angle $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube. Prove that $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = 4/3$.

Soln:



Let OCMANBLP be a cube whose edge is a . The coordinates of the vertices are respectively $O(0,0,0)$, $A(a,0,0)$, $B(0,a,0)$, $C(0,0,a)$, $M(a,0,a)$, $N(a,a,0)$, $L(0,a,a)$, $P(a,a,a)$;

Now,

$$\begin{aligned} OP &= \sqrt{\{(0-a)^2 + (0-a)^2 + (0-a)^2\}} \\ &= \sqrt{a^2 + a^2 + a^2} \\ &= \sqrt{3}a \end{aligned}$$

Here, the four diagonals of the cube OP , AL , BM , CN are equal.

i.e., $OP = AL = BM = CN = \sqrt{3} a$

The dcs of the diagonal OP are:

$$\frac{a-0}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a}$$

i.e., $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

The dcs of AL are:

$$\frac{0-a}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a}$$

i.e., $-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

The dcs of BM are:

$$\frac{a-0}{\sqrt{3}a}, \frac{0-a}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a}$$

i.e., $\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

The dcs of CN are:

$$\frac{a-0}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a}, \frac{0-a}{\sqrt{3}a}$$

i.e., $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}$

let l, m, n be the direction cosines of the line that makes angle $\alpha, \beta, \gamma, \delta$ with the four diagonals OP, AL, BM and CN , respectively,

Then,

$$\cos \alpha = l \cdot \left(\frac{1}{\sqrt{3}}\right) + m \cdot \left(\frac{1}{\sqrt{3}}\right) + n \cdot \left(\frac{1}{\sqrt{3}}\right) = \frac{l+m+n}{\sqrt{3}} \quad \text{--- (1)}$$

$$\cos \beta = l \left(-\frac{1}{\sqrt{3}}\right) + m \left(\frac{1}{\sqrt{3}}\right) + n \left(\frac{1}{\sqrt{3}}\right) = \frac{-l+m+n}{\sqrt{3}} \quad \text{--- (2)}$$

$$\cos \gamma = l \left(\frac{1}{\sqrt{3}}\right) + m \left(-\frac{1}{\sqrt{3}}\right) + n \left(\frac{1}{\sqrt{3}}\right) = \frac{l-m+n}{\sqrt{3}} \quad \text{--- (3)}$$

$$\cos \delta = l \left(\frac{1}{\sqrt{3}}\right) + m \left(\frac{1}{\sqrt{3}}\right) + n \left(-\frac{1}{\sqrt{3}}\right) = \frac{l+m-n}{\sqrt{3}} \quad \text{--- (4)}$$

Now, squaring & adding (1), (2), (3) & (4) we get.

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{1}{3} \left[(l+m+n)^2 + (-l+m+n)^2 + (l-m+n)^2 + (l+m-n)^2 \right]$$

$$= \frac{1}{3} \left[l^2 + m^2 + n^2 + 2lm + 2mn + 2nl + l^2 + m^2 + n^2 - 2lm + 2mn - 2nl + l^2 + m^2 + n^2 - 2nl - 2mn + 2nl + l^2 + m^2 + n^2 + 2lm - 2mn - 2ln \right]$$

$$= \frac{4}{3} (l^2 + m^2 + n^2) = \frac{4}{3} (1) \quad [\because l^2 + m^2 + n^2 = 1]$$

$$\therefore \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}$$

Proved

Corollary° prove that $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma + \sin^2 \delta = 8/3$

We know, $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = 4/3$

$$\Rightarrow 1 - \sin^2 \alpha + 1 - \sin^2 \beta + 1 - \sin^2 \gamma + 1 - \sin^2 \delta = 4/3$$

$$\Rightarrow \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma + \sin^2 \delta = 4 - 4/3 =$$

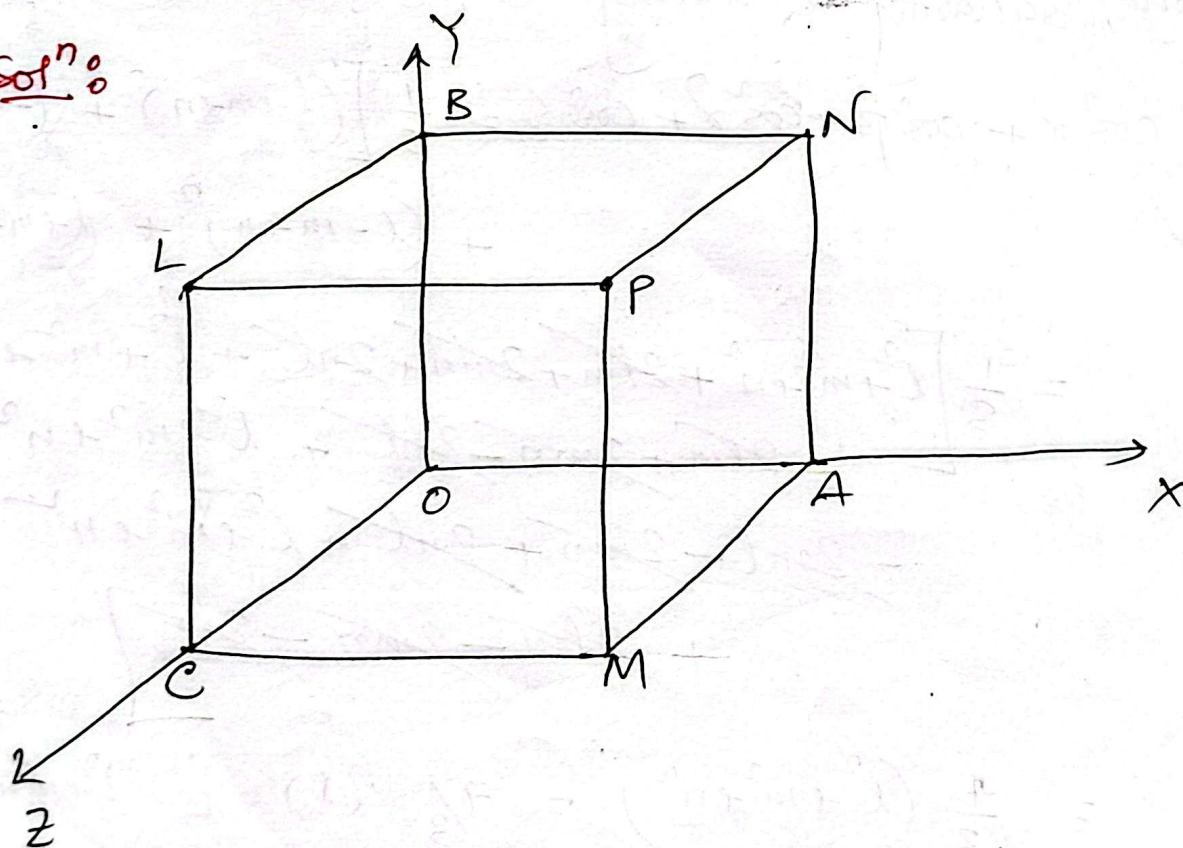
$$\therefore \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma + \sin^2 \delta = 8/3$$

(Proved)

Problem°

Prove that the angle betⁿ two diagonals of a cube is $\cos^{-1}(1/3) \approx 70.53^\circ$

Solⁿ°



Let OCMANBLP be a cube whose edge is a . The coordinates of the vertices are respectively,

$$O(0,0,0), A(a,0,0), B(0,a,0), C(0,0,a), M(a,0,a), N(a,a,0), L(0,a,a), P(a,a,a);$$

The dir^s of the diagonal OP is:

$$a-0, a-0, a-0$$

$$\text{i.e., } a, a, a$$

The dir^s of the diagonal AL is

$$0-a, a-0, a-0$$

$$\text{i.e., } -a, a, a$$

Let, θ be the angle betⁿ OP and AL

$$\therefore \cos \theta = \frac{a(-a) + a \cdot a + a \cdot a}{\sqrt{a^2 + a^2 + a^2} \cdot \sqrt{a^2 + a^2 + a^2}}$$

$$= \frac{-a^2 + a^2 + a^2}{\sqrt{3}a \cdot \sqrt{3}a}$$

$$\Rightarrow \cos \theta = \frac{a^2}{3a^2}$$

$$\Rightarrow \cos \theta = \frac{1}{3}$$

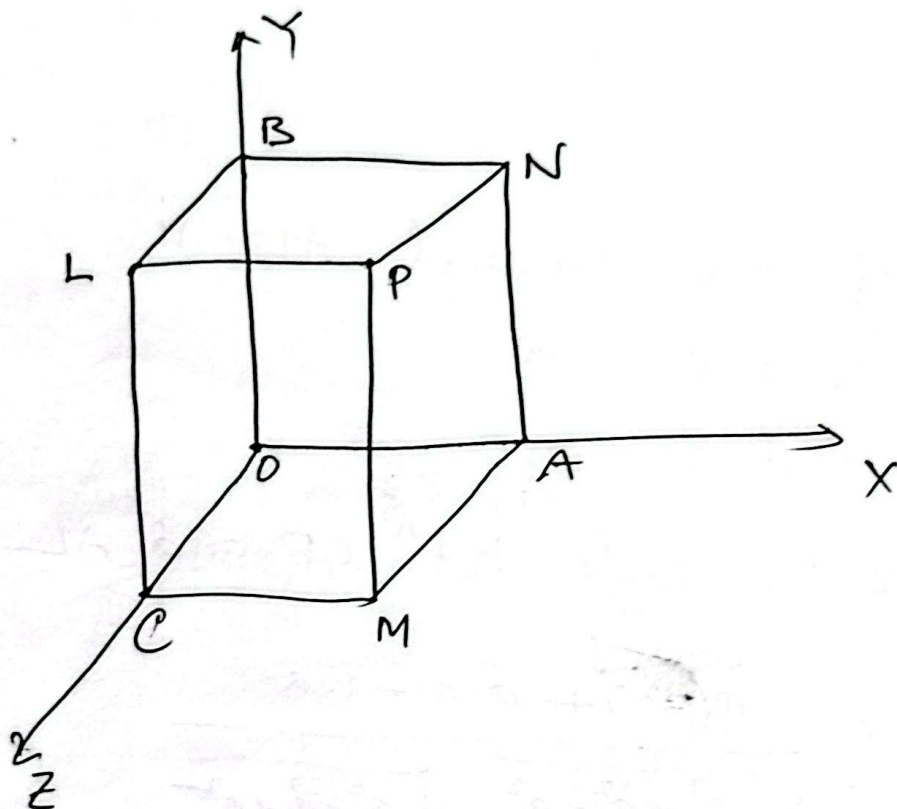
$$\therefore \theta = \cos^{-1}\left(\frac{1}{3}\right) \approx 70.53^\circ$$

(Proved)

Problem:

If the edges of a rectangular parallelepiped are a, b, c , show that the angle between the four diagonals are given by $\cos^{-1}\left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2}\right)$

Solⁿ:



Take one of the vertices O of the parallelepiped as origin and three rectangular faces through it as the three rectangular coordinate planes.

Let $OA = a$, $OB = b$, $OC = c$.

The lines OP, AL, BM and CN are four diagonals of the parallelepiped. whose coordinates are $O(0, 0, 0)$, $A(a, 0, 0)$, $B(0, b, 0)$, $C(0, 0, c)$, $M(a, 0, c)$

$N(a, b, 0)$, $L(0, b, c)$ and $P(a, b, c)$

$$\text{DC}^s \text{ of } OP: \frac{a-0}{\Sigma a^2}, \frac{b-0}{\Sigma a^2}, \frac{c-0}{\Sigma a^2}$$
$$\text{i.e., } \frac{a}{\Sigma a^2}, \frac{b}{\Sigma a^2}, \frac{c}{\Sigma a^2}$$

$$\text{DC}^s \text{ of } AL: \frac{0-a}{\Sigma a^2}, \frac{b-0}{\Sigma a^2}, \frac{c-0}{\Sigma a^2}$$
$$\text{i.e., } \frac{-a}{\Sigma a^2}, \frac{b}{\Sigma a^2}, \frac{c}{\Sigma a^2}$$

$$\text{DC}^s \text{ of } BM: \frac{a-0}{\Sigma a^2}, \frac{0-b}{\Sigma a^2}, \frac{c-0}{\Sigma a^2}$$
$$\text{i.e., } \frac{a}{\Sigma a^2}, \frac{-b}{\Sigma a^2}, \frac{c}{\Sigma a^2}$$

$$\text{DC}^s \text{ of } CN: \frac{a-0}{\Sigma a^2}, \frac{b-0}{\Sigma a^2}, \frac{0-c}{\Sigma a^2}$$
$$\text{i.e., } \frac{a}{\Sigma a^2}, \frac{b}{\Sigma a^2}, \frac{-c}{\Sigma a^2}$$

Let the angle betⁿ OP & AL is θ

$$\therefore \cos \theta = \frac{a(-a) + b(b) + c(c)}{\sqrt{\Sigma a^2} \cdot \sqrt{\Sigma a^2}}$$

$$\Rightarrow \cos \theta = \frac{-a^2 + b^2 + c^2}{a^2 + b^2 + c^2}$$

$$\therefore \theta = \cos^{-1} \left(\frac{-a^2 + b^2 + c^2}{a^2 + b^2 + c^2} \right).$$

Similarly, the angle between any one of the six pairs of diagonals can be found, as

Hence, the angles, in general, betⁿ the six pairs ^{${}^4C_2 = 6$} are given by,

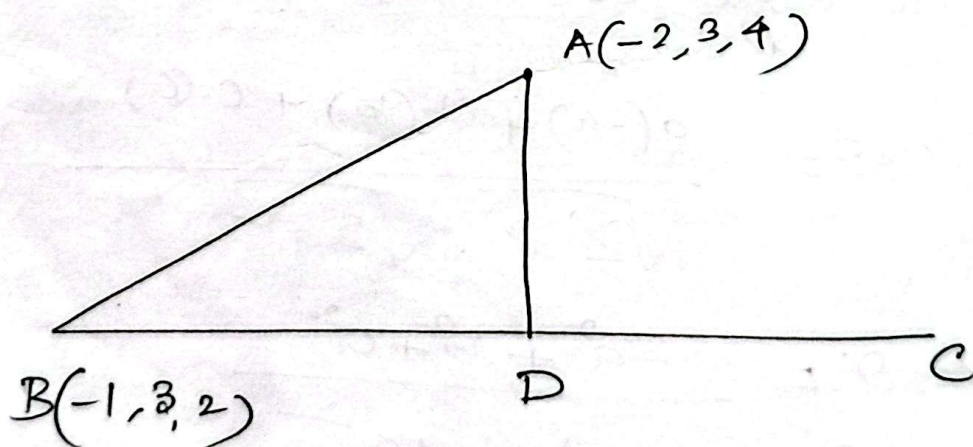
$$\cos^{-1} \left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2} \right)$$

(Proved)

Problem:

Find the distance of $(-2, 3, 4)$ from the line through the point $(-1, 3, 2)$ where the d.c's are proportional to $12, 3, -4$.

Solⁿ:



Let BC be the line passing through the point $B(-1, 3, 2)$ whose d.c's are proportional to $12, 3, -4$
 Let, l, m, n be the d.c's of BC

$$\frac{l}{12} = \frac{m}{3} = \frac{n}{-4} = \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{(12)^2 + (3)^2 + (-4)^2}} = \frac{1}{\sqrt{169}} = \frac{1}{13}$$

$$\therefore l = \frac{12}{13}, m = \frac{3}{13}, n = -\frac{4}{13}$$

$$\therefore AB = \sqrt{(-2+1)^2 + (3-3)^2 + (4-2)^2} = \sqrt{1+0+4} = \sqrt{5}$$

Draw $AD \perp BC$ from the point A.

BD = The projection BA on BC

$$\Rightarrow BD = \frac{12}{13}(-2+1) + \frac{3}{13}(3-3) + \left(-\frac{4}{13}\right)(4-2)$$

$$= -\frac{12}{13} + 0 - \frac{8}{13}$$

$$= -\frac{20}{13}$$

From the right angled triangle ABD

$$AB^2 = AD^2 + BD^2$$

$$\Rightarrow AD = \sqrt{AB^2 - BD^2}$$

$$= \sqrt{(\sqrt{5})^2 - \left(-\frac{20}{13}\right)^2}$$

$$\Rightarrow AD = \sqrt{5 - \frac{400}{169}}$$

$$= \frac{\sqrt{845 - 400}}{\sqrt{169}}$$

$$\therefore AD = \frac{\sqrt{445}}{13}$$

Example:

Show that the equation of a right circular cone whose vertex is at the origin, whose axes have d.c's $\cos \alpha, \cos \beta, \cos \gamma$, and semi-vertical angle is θ is given by

$$(x \cos \beta - y \cos \alpha)^2 + (y \cos \gamma - z \cos \beta)^2 + (z \cos \alpha - x \cos \gamma)^2 = (x^2 + y^2 + z^2) \sin^2 \theta$$

Soln:

Let $P(x, y, z)$ be any point on the right circular cone. Take the circular section of the cone through the point P . Let C be the center of the circle.

Then $PC \perp OC$

$$\therefore \sin \theta = \frac{PC}{OP}$$

$$\Rightarrow PC^2 = OP^2 \sin^2 \theta \quad \text{--- (1)}$$

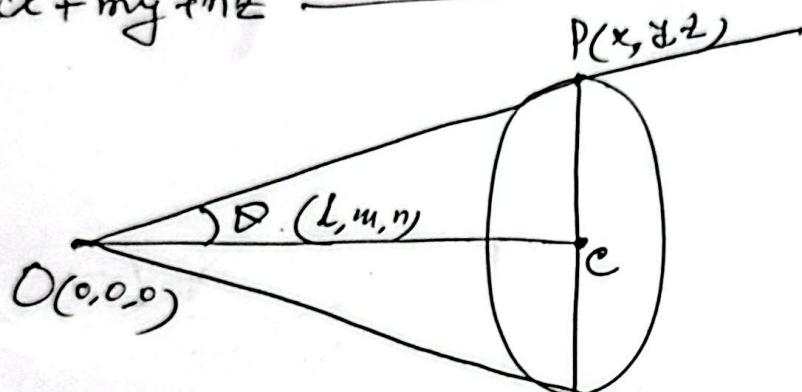
$$OP^2 = x^2 + y^2 + z^2 \longrightarrow (11)$$

$OC = \text{projection of } OP \text{ on } OC$

$$= (x-0)\cos\alpha + (y-0)\cos\beta + (z-0)\cos\gamma$$

$$\Rightarrow OC = x\cos\alpha + y\cos\beta + z\cos\gamma \quad (14)$$

$$= lx + my + nz$$



Now, using Pythagorean th^m on the right-angled triangle OPC we have

$$OP^2 = OC^2 + PC^2$$

$$\Rightarrow PC^2 = OP^2 - OC^2$$

$$\Rightarrow PC^2 = (x^2 + y^2 + z^2) - (lx + my + nz)^2$$

$$\Rightarrow PC^2 = (x^2 + y^2 + z^2)(l^2 + m^2 + n^2) - (lx + my + nz)^2$$

$$= (yn - mz)^2 + (zL - nz)^2 + (xm - ly)^2$$

$$\Rightarrow PC^2 = (y\cos\gamma - z\cos\beta)^2 + (z\cos\alpha - x\cos\gamma)^2 + (x\cos\beta - y\cos\alpha)^2 \quad [L. Identity]$$

$$\Rightarrow OP^2 \sin^2 \theta = (x\cos\beta - y\cos\alpha)^2 + (y\cos\gamma - z\cos\beta)^2 + (z\cos\alpha - x\cos\gamma)^2$$

$$\Rightarrow OP^2 (x^2 + y^2 + z^2) \sin^2 \theta = DO$$